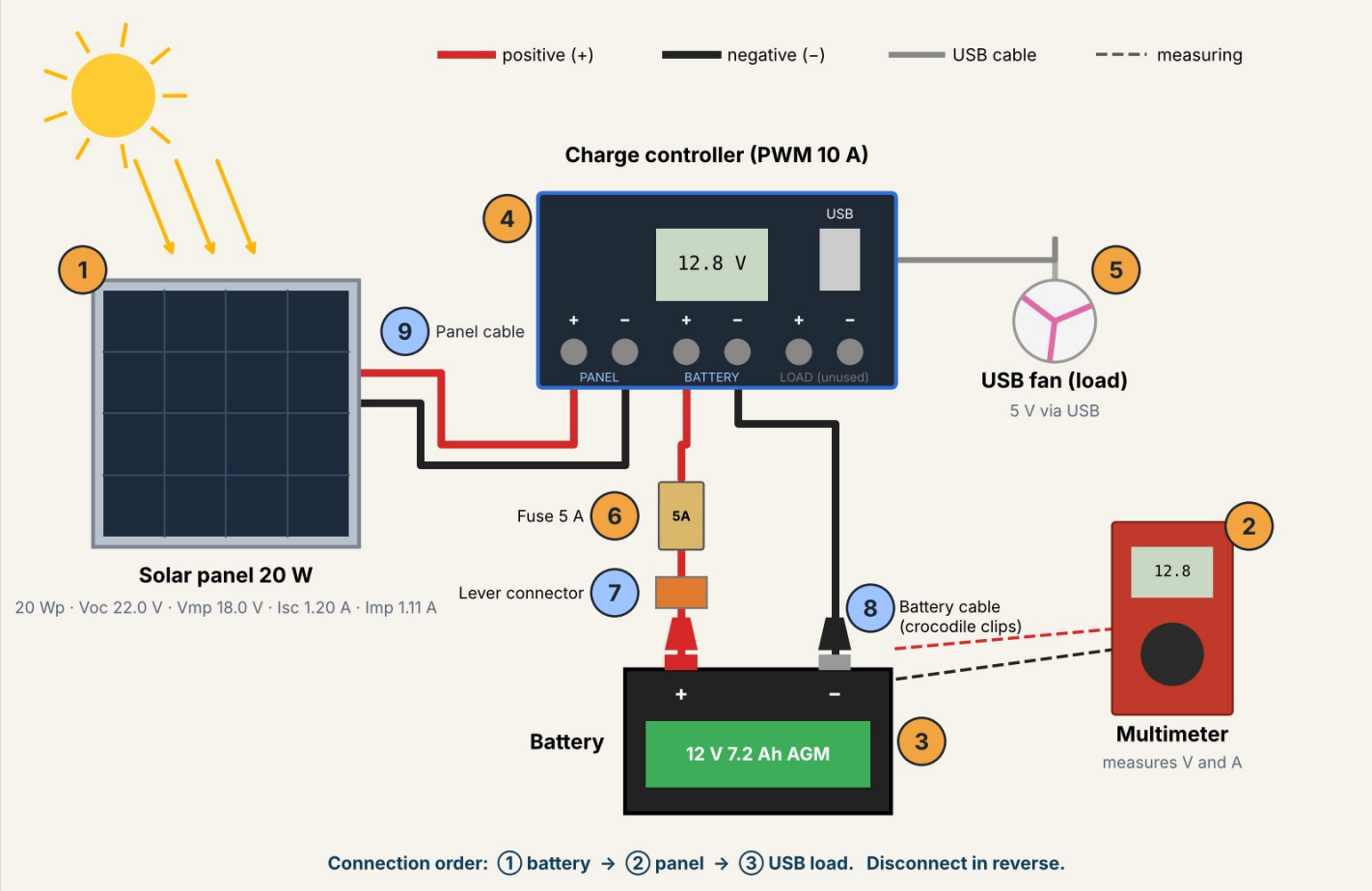
# Solar + Battery Test System — Readings Log

Mohamed Eniw · Passau, Germany · October 2026 · values measured with an AstroAI AM33D multimeter

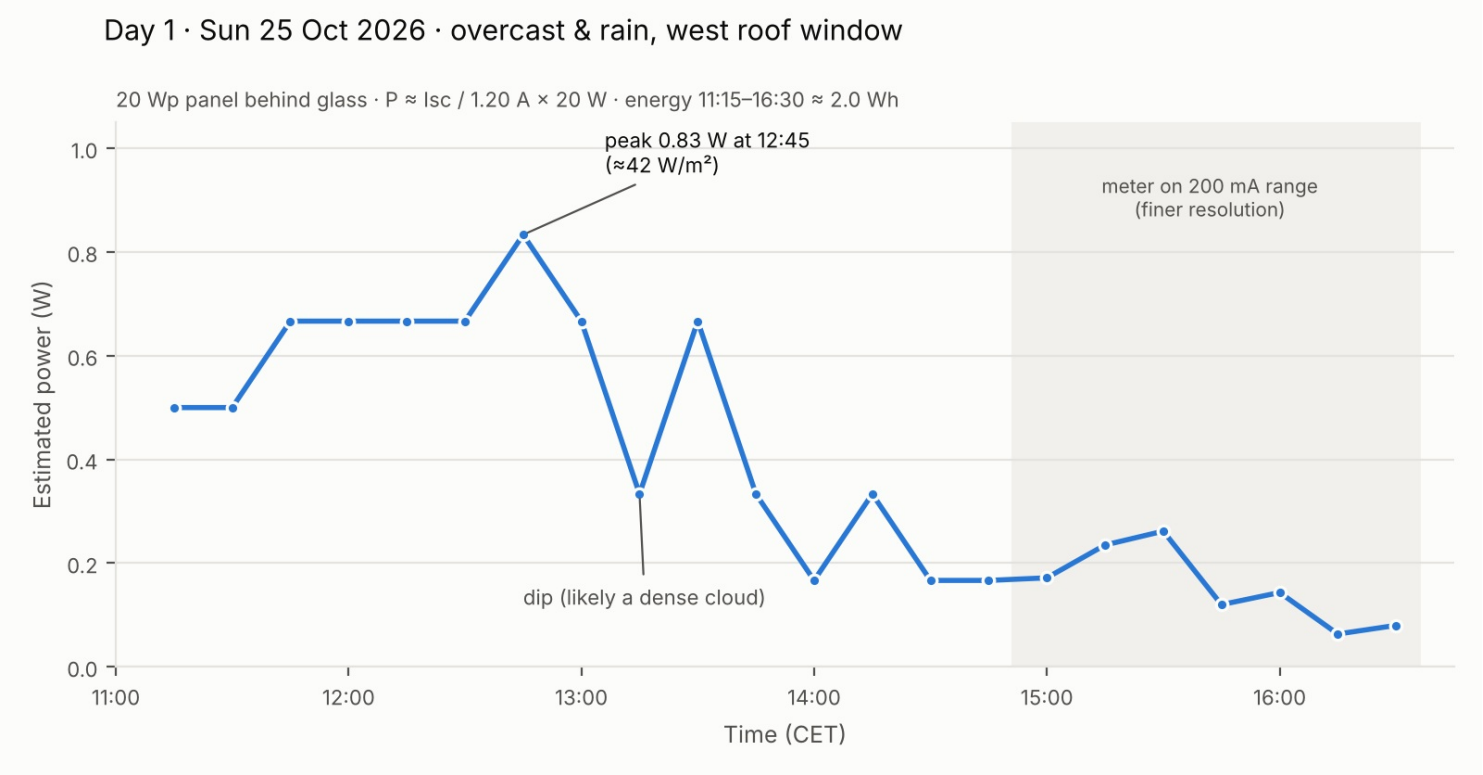
| # | Measurement | Parts | Meter setting | Expected | Measured | Time / conditions |
|---|---|---|---|---|---|---|
| R1 | Battery voltage at rest (before wiring) | 2 on 3 | V⎓ 20 | 12.6–13.0 V | **12.94 V** | Sat 24 Oct, 18:01, indoors |
| R2 | Battery voltage shown on controller | 4 | display | ≈ R1 | **12.7 V** | Sat 24 Oct, 19:22, indoors |
| R3 | Battery voltage with fan running | 2 on 3 | V⎓ 20 | slightly below R1 | **12.68 V** | Sat 24 Oct, 19:29, indoors |
| R4 | Panel open-circuit voltage (Voc), sun | 2 on 1 | V⎓ 200 | ≈ 20–22.0 V (label Voc 22.0 V) | **21.8 V** | Sun 25 Oct, 10:54, behind window, overcast/rain |
| R5 | Panel short-circuit current (Isc), sun | 2 on 1 | A 10 (10 A socket) | ≤ 1.20 A (label Isc) | **0.04 A** | Sun 25 Oct, 10:59, behind window, overcast/rain |
| R6 | Battery voltage while charging | 2 on 3 | V⎓ 20 | rises above R1 | | |
| R7 | Charging current panel → controller | 2 in 9 | A 10 (10 A socket) | up to ≈ 1 A | | |

**Parts:** 1 Solar panel 20 W · 2 Multimeter · 3 Battery 12 V 7.2 Ah AGM · 4 Charge controller PWM 10 A · 5 USB fan · 6 Fuse 5 A · 7 Lever connector · 8 Battery cable · 9 Panel cable



## Daily production log

Short-circuit current (Isc) logged by camera every 15 min · panel fixed behind a west-facing roof window (284°) · P ≈ Isc / 1.20 A × 20 W (estimate)




| Day | Weather | Peak | Energy (logged window) |
|---|---|---|---|
| Sun 25 Oct | Overcast, rain | 0.05 A at 12:45 (≈ 0.83 W, ≈ 42 W/m²) | ≈ 2.0 Wh (11:15–16:30) |
| Mon 26 Oct | – | – | – |
| Tue 27 Oct | – | – | – |